

Roll No.

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Paper Code: TEE-201

B.Tech (EEE), End-Semester BACK Examination 2019

II Semester

Basic Electrical Engineering

Time: Three Hours

MM: 100

Note:

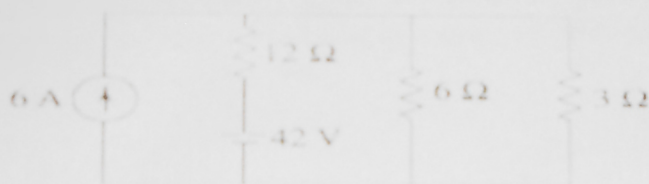
- The question paper contains five questions.
- All questions are compulsory.
- Instructions on how to attempt a question are mention against it.
- Total marks assigned to each question are twenty.

Q1. (Attempt any two questions of choice a, b and c)

a. State Thevenin's theorem and explain this theorem with steps from any example.

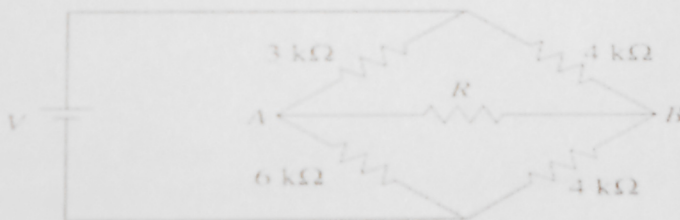
b.

Find the current through the  $12\Omega$  resistor.



c.

The value of the resistance  $R$  connected across the terminals  $A$  and  $B$ , which will absorb the maximum power is



Q2. (Attempt any two questions of choice a, b and c)

a. Define the following (i) active and passive element (ii) Bandwidth factor (iii) rms and average value (iv) time period and frequency.

b. A coil having resistance of 20 ohm and an inductance of 0.6 H is connected in series with a condenser of capacitance 20 mF. A voltage of 250 volts at variable frequency is applied to the combination. At what frequency, will the current be maximum? Calculate this current and find quality factor and bandwidth of this circuit.

c. Derive the condition of resonance in series RLC circuit with graphical representation and also derive expression for quality factor in series RLC circuit.

Q3. (Attempt any two questions of choice a, b and c)

- a. (i) What is the principle of transformer?  
 (ii) What is the analogy between electrical and magnetic circuit?  
 (iii) Define different testing mode on transformer.
- b. The power into a balanced 3-phase inductive load measured by two wattmeter's are 2500 W and 1500 W. The line voltages are 440 V, 50 Hz. Determine the active power, reactive power, kVA and power factor of the load.
- c. Explain all type of circuits designed on DC machine during its operation at full load.

Q4. (Attempt any two questions of choice a, b and c)

- (a) Prove that active and reactive power in three phase balanced star circuit can be measured by two Wattmeter method.
- (b)  
 A 40 kVA transformer has iron loss of 350W and full-load copper loss of 450W. If power factor of the load is 0.8 lagging, calculate (i) full-load efficiency, (ii) the load at which maximum efficiency occurs, and (iii) the maximum efficiency.
- (c) Define Maximum power transfer theorem for dc circuit and derive the condition of maximum efficiency obtained.

Q5. (Attempt any two questions of choice a, b and c)

- (a) (i) What is the principle of DC machine explain with its construction diagram.
- (b)  
*The armature of a four-pole, lap-wound shunt generator has 120 slots with 4 conductors per slot. The flux per pole is 0.05 Wb. The armature resistance is 0.05  $\Omega$  and the shunt-field resistance is 50  $\Omega$ . Find the speed of the machine when supplying 450 A at a terminal voltage of 250 V.*
- (c) The power input to the rotor of a 480-V, 60Hz, 6-pole, 3-phase induction motor is 150KW. The rotor electromotive force is observed to make at 1500 rpm, Calculate (i) the slip (ii) the rotor speed (iii) speed of stator field with respect to rotor (iv) Frequency of rotor currents at standstill (v) speed of rotor magnetic field with respect to stator magnetic field.